Grupo Dados | Encontro 01 | Data 11/04/2026 | A.M.D.G

code names.py

```
name = input(...)
print(f"...")
```

```
names = []
for _ in range(3):
    name = input(...)
    names.append(name)
```

↳ for name in sorted(names)
    print(f'...')

↳ ⊛ Perda da organização

open (double click)

```
name = input(...)
file = open("names.txt", "w")
file.write(name)
file.close()
```

"a"

↳ f"{name}\n"

with

↳ with open("...", "a") as file:
    file.write(...)

→ Leitura

code leitura_names.py

```
with open("names.txt", "r") as file:
    line = file.readlines()

for line in lines:
    print("hello,", line)
```

```
names = []
:
names.append(line)

for name in sorted(names):
    print ...
```

code ~~nomes~~ alunos.csv

"comma separated values"

Gildo, ...
Fapesp, ...
Brasfemp ...

code alunos.py

```
with open("alunos.csv") as file:
    for line in file:
        row = line.rstrip().split(",")
        print(row[0], row[1])
```

→ nome, curso

→ Para ordenarmos...

```
alunos = [...]
    ⋮
aluno = {"nome": ..., "curso": ...}
    ⋮
def curso_aluno(aluno):
    return aluno["curso"]
```

```
for aluno in sorted(
    alunos, key = curso_aluno):
    print(...)
```

atenção

lambda

```
key = lambda student:
        student["nome"]
```

import csv

Alterar no CSV: nome, curso

```
alunos = []
with open(...) as file:
    reader = csv.reader(file)
    for row in reader:
        alunos.append({...})
```

DictReader

row["nome"]
row[0], row[1]

code escrita_csv.py

```
nome = ...
curso = ...

with open("alunos.csv", "a") as file:
    writer = csv.DictWriter(file, fieldnames =
                                    ["nome", "curso"])
    writer.writerow({"curso": curso,
                     "nome": nome})
```

code criação_gif.py (PIL)

```
import sys

from PIL import image

images = [ ]
for arg in sys.argv[1:]:

    image = Image.open(arg)
    images.append(image)

imagens[0].save("kirin.gif",
                save_all = True,
                append_images = [imagens[1]],
                duration = 1000,
                loop = 0
                )
```

Grupo Dados | Encontro 08 | Data 02/05/2016

code teste_email.py

```
email = input(...).strip()

if "@" in email:
    print(valido!) and "." in email

else:
    print("formato incorreto!")
```

```
user, domain = email.split("@")

if (username) and (":" in domain):
    print(...)

else:
    print(...)
```

Upgrade:

and domain.endswith(".edu")

↳ gilbo@usp.com.br

re

re.search(pattern, string, flags=0)

```
import re

if re.search("@", email):
    print("...")

else:
    print("...")
```

(*)

- . any character except a newline
- * 0 or more repetitions
- + 1 or more repetitions
- ? 0 or 1 repetition
- {m} m repetitions
- {m,n} m-n repetitions

↳ "@" ⇝ ".*@.*" ⇝ ".+@.+"

⇝ ".+@.+.com"

↳ meu email é gilbo@usp.com

⊛ ^ matches the start of the string

$ matches the end of the string

↳ r"^. + @. + \.com $"

⊛ [ ] set of characters

[^] complementing the set

↳ r"^ [^@] + @ [^@] + \.com $"

[a-zA-Z0-9_]

↳ \w

⊛ \d decimal digit

\D not " "

\s whitespace character

\S not " "

\w word character

\W not " "

A|B either A or B

(...) a group

57min